

CONTENTS

FOREWORD.....	5
1 Scope.....	7
2 Normative references.....	8
3 Terms and definitions	8
4 Symbols, subscripts and superscripts.....	13
4.1 General.....	13
4.2 Symbols.....	13
4.3 Subscripts.....	15
4.4 Superscripts.....	16
5 Characteristics of short-circuit currents: calculating method.....	16
5.1 General.....	16
5.2 Calculation assumptions.....	19
5.3 Method of calculation	20
5.3.1 Equivalent voltage source at the short-circuit location	20
5.3.2 Symmetrical components	22
6 Short-circuit impedances of electrical equipment	23
6.1 General.....	23
6.2 Network feeders	23
6.3 Transformers	25
6.3.1 Two-winding transformers	25
6.3.2 Three-winding transformers.....	25
6.3.3 Impedance correction factors for two- and three-winding network transformers	27
6.4 Overhead lines and cables	28
6.5 Short-circuit current-limiting reactors	29
6.6 Synchronous machines	29
6.6.1 Synchronous generators	29
6.6.2 Synchronous compensators and motors.....	31
6.7 Power station units.....	31
6.7.1 Power station units with on-load tap-changer.....	31
6.7.2 Power station units without on-load tap-changer	32
6.8 Wind power station units	33
6.8.1 General	33
6.8.2 Wind power station units with asynchronous generator	33
6.8.3 Wind power station units with doubly fed asynchronous generator.....	34
6.9 Power station units with full size converter.....	35
6.10 Asynchronous motors.....	35
6.11 Static converter fed drives.....	36
6.12 Capacitors and non-rotating loads	36
7 Calculation of initial short-circuit current.....	36
7.1 General.....	36
7.1.1 Overview	36
7.1.2 Maximum and minimum short-circuit currents	41
7.1.3 Contribution of asynchronous motors to the short-circuit current.....	42
7.2 Three-phase initial short-circuit current.....	43
7.2.1 General	43

7.2.2	Short-circuit currents inside a power station unit with on-load tap-changer	44
7.2.3	Short-circuit currents inside a power station unit without on-load tap-changer	46
7.3	Line-to-line short circuit	47
7.4	Line-to-line short circuit with earth connection	47
7.5	Line-to-earth short circuit	49
8	Calculation of peak short-circuit current	49
8.1	Three-phase short circuit	49
8.1.1	Single-fed and multiple single-fed short circuits	49
8.1.2	Multiple-fed short circuit	51
8.2	Line-to-line short circuit	52
8.3	Line-to-line short circuit with earth connection	52
8.4	Line-to-earth short circuit	52
9	Calculation of symmetrical breaking current	53
9.1	Three-phase short circuit	53
9.1.1	Symmetrical breaking current of synchronous machines	53
9.1.2	Symmetrical breaking current of asynchronous machines	54
9.1.3	Symmetrical breaking current of power station units with doubly fed asynchronous generator	55
9.1.4	Symmetrical breaking current of power station units with full size converter	55
9.1.5	Symmetrical breaking current of network feeder	56
9.1.6	Symmetrical breaking current in case of multiple single-fed short-circuits	56
9.1.7	Symmetrical breaking current in case of multiple-fed short circuits	56
9.2	Unbalanced short-circuits	57
10	DC component of the short-circuit current	57
11	Calculation of steady-state short-circuit current	58
11.1	General	58
11.2	Three-phase short circuit	58
11.2.1	Steady-state short-circuit current of one synchronous generator or one power station unit	58
11.2.2	Steady-state short-circuit current of asynchronous motor or generator	61
11.2.3	Steady-state short-circuit current of wind power station unit with doubly fed asynchronous generator	61
11.2.4	Steady-state short-circuit current of wind power station unit with full size converter	61
11.2.5	Steady-state short-circuit current of network feeder	61
11.2.6	Steady-state short-circuit current in case of multiple single-fed short circuits	61
11.2.7	Steady-state short-circuit current in case of multiple-fed short circuits	62
11.3	Unbalanced short circuits	62
12	Short circuits at the low-voltage side of transformers, if one line conductor is interrupted at the high-voltage side	62
13	Terminal short circuit of asynchronous motors	64
14	Joule integral and thermal equivalent short-circuit current	65
Annex A (normative)	Formulas for the calculation of the factors m and n	68
Annex B (informative)	Nodal admittance and nodal impedance matrices	69
Bibliography		73